

Accuracy of the Scaled Flux Method for Spatially Resolved Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors

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Review General Response

We would like to thank the reviewers for their detailed assessment of this paper. We have fixed many issues, updated simulation parameters, and generally improved the paper thanks to your suggestions.

Reviewer 1

1. The distinction between conditions where the scaled flux method holds versus fails is well argued, and it clarifies the applicability of a commonly used approximation.

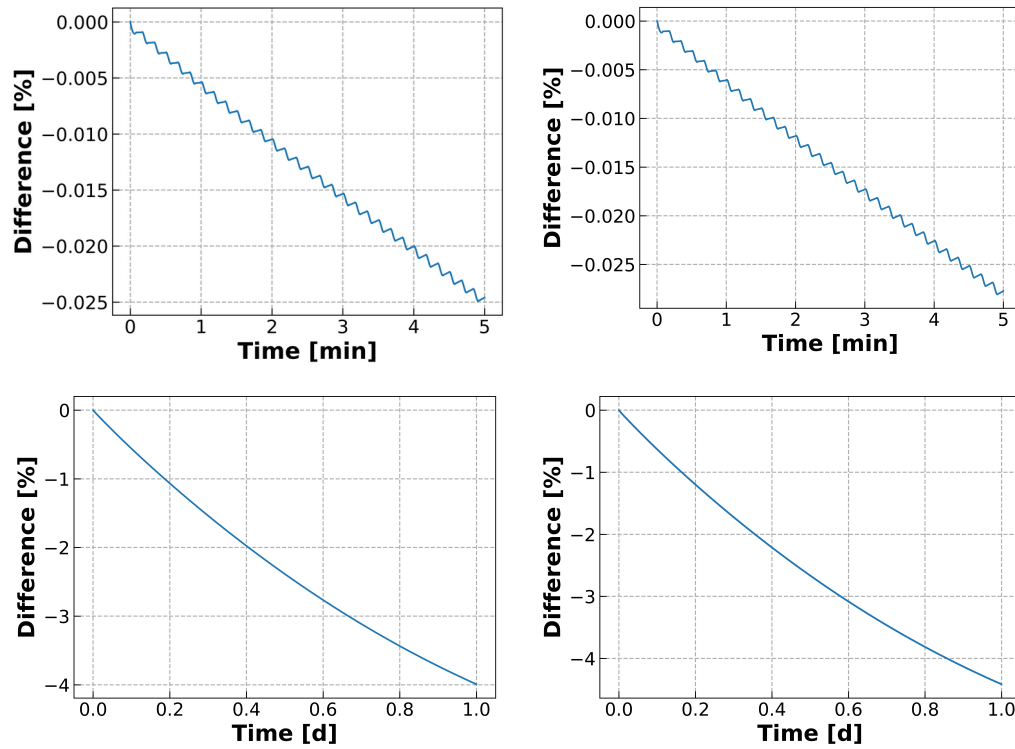
Solution: Thank you!

2. The manuscript states that simulations performed for MSRE primary loop where the temperature is approximately 911K. However, according to the simulation parameters listed in TABLE I, authors set the temperature at 294K, which is extremely smaller than a typical molten salt reactor conditions. Therefore, the temperature chosen for the simulations seem inconsistent with the application and may significantly affect results and the conclusion made based on them. It is recommended to reconsider the temperature choice or add more explanation for the justification.

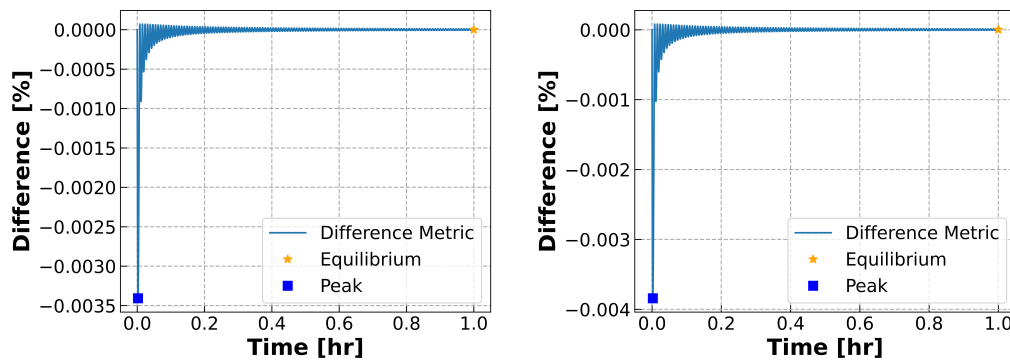
Solution: Thank you for pointing this out! A more appropriate temperature of 900K is now used in Table I. The difference in the fission source term for ^{135}Xe , $\phi(t)FY_i\Sigma_f$, is 0.075%, while the absorption term difference, $\phi(t)\sigma_{a,i}$, is 1.4%. Based on these differences, the results are recalculated, with the following changes:

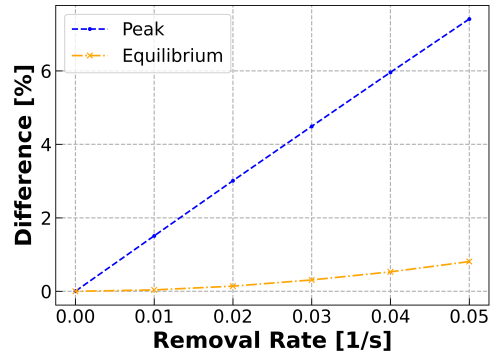
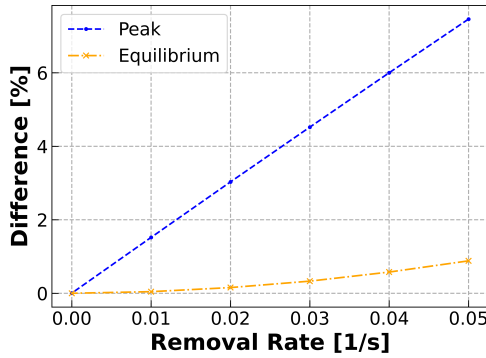
- Table II - The general trends remain the same, but slight changes to the values.
- Table III - Minimal differences with same general trend.
- Table IV and V - The general trends remain the same, but slight changes to the values.

- Table VI - The general trends remain the same, but the differences change. All differences decreased except for the Two Region ^{135}Xe and the Two Region ^{135}I .
- Figure 4 - The plots generally remains the same, but there is a slight difference in the resulting values (original on the left, updated on the right):



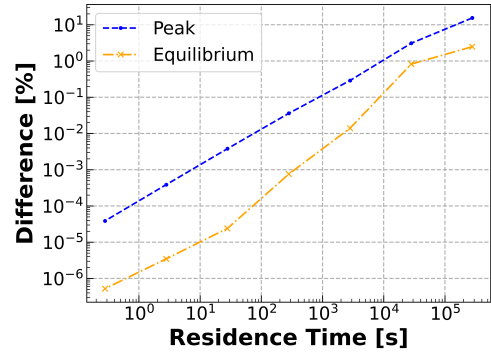
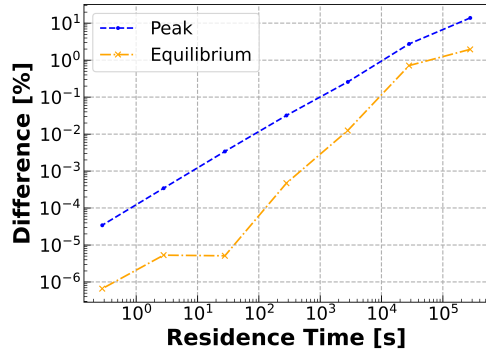
- Figure 5 - The plots generally remains the same, but there is a slight difference in the resulting values (original on the left, updated on the right):





The difference is negligible in the figure plotting over removal rate because the small differences introduced by changing the temperature are negligible compared to the large chemical removal rate.

- Figure 6 - The plots generally remains the same, but there is a slight difference in the resulting values (original on the left, updated on the right):



- Validation against MSRE data is valuable but limited by experimental uncertainties and chemical effects not modeled. This limitation is acknowledged, but the discussion on uncertainties in the measurements could be deepened. In reference to statement: "The lack of agreement for ⁹⁵Zr and ⁹⁵Nb ..." (Page 13-line 57), how can we validate the methods if there is no agreement with MSRE experiments? Please explain.

Solution: This phrasing was unclear, thank you for noticing. The paragraph was restructured as follows:

However, the MSRE experimental data does not agree at each point, particularly for ⁹⁵Nb. The ⁹⁵Nb concentrations from the MSRE measurements are smaller than the model predicts, which means that a non-zero chemical removal rate would likely fit these data better. This lack of agreement can be explained by uncertainty in the measurements, chemical effects which aren't modeled in MCNP/ADDER or this work, and other contributing factors

these models do not include.

The MSRE experimental data for ^{95}Zr agrees reasonably well with the models presented in this work, as the points generally fall on or near the modeled concentration. These models are not expected to perfectly capture the physics of the MSRE, but rather the general trend of the evolution of the nuclide concentrations. Thus, the observed agreement with ^{95}Zr demonstrates the models are successfully capturing this general trend.

4. There are a few typographical errors, and they should be fixed. (e.g. page 4-line 50: "an laptop" → "a laptop", page 18-line 45: "... and than" → "... and then", etc.)

Solution: Thank you for pointing these out, they have been corrected.

5. The reference needs to be added for the information given on Page 3-line 20: "... , such as the 2020 Advanced Reactor Demonstration Program award for \$113 million over seven years to the Molten Chloride Reactor Experiment."

Solution: Added citation [1].

6. Page 6-line 33: "... the neutron total capture cross section ..." needs to be replaced with "... the total neutron capture cross-section ..." for clarity.

Solution: This line has been corrected.

7. It would have been better if we see a geometry scheme for 1D representation of MSRE primary loop for a better visualization of spatial nodes.

Solution: This is a good point. The following paragraph and figure have been added:

Figure 1 shows the flow path along which the spatial nodes are distributed. Most of the path is in the ex-core region, but 27.2% is in the in-core region, which takes up 165.39 centimeters of the 608.06 centimeter flow path length.

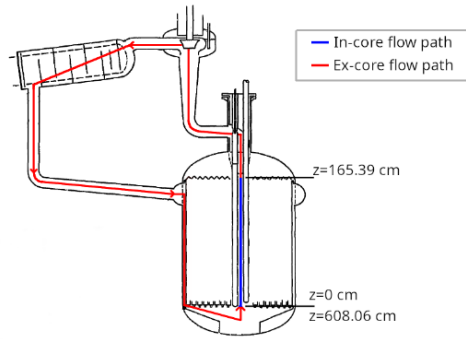


Fig. 1. The flow path and bounds for the in-core region of the MSRE, overlaid on the MSRE loop diagram from [14].

8. To increase the readability, it is recommended to add a bold highlight for the "500 nodes" in Table II.

Solution: This is a great suggestion! I have added this bold highlight to each table used in the parameter sensitivity studies to note which values are used for the difference sensitivity studies. Additionally, since the Courant number of 1.0 is used for most of the other analyses, these sensitivity studies now also use that Courant number. These updated tables are included here:

TABLE II

Spatial sensitivity detailing how cost and accuracy vary based on the number of spatial nodes for ^{135}Xe with a single tracked nuclide and a Courant number of 1.0 over one hour in the MSRE. The number of nodes selected for generating results is shown in bold.

Nodes	Cost [s]	Error [%]
50	0.14	2.59
100	0.42	1.25
200	1.29	0.82
500	6.93	0.11
1000	26.63	0.06
2000	103.12	0.01
4000	413.30	-

TABLE III

Chain sensitivity detailing how cost and accuracy vary based on the number of nuclides for ^{135}Xe with 500 spatial nodes and a Courant number of 1.0 over one hour in the MSRE. The nuclides selected for generating results are shown in bold.

Nuclides	Cost [s]	Error [%]
Xe	7.40	84.05
Xe, ^mXe	15.01	58.38
Xe, ^mXe , I	21.27	31.73
Xe, ^mXe , I, Te	28.10	1.08
Xe, ^mXe, I, Te, Sb	35.30	0.00
Xe, ^mXe , I, Te, Sb, Sn	42.14	-

TABLE IV

Courant number sensitivity study on the concentration of ^{135}Xe with five nuclides in the MSRE based on the time step size used with 500 spatial nodes over the course of one minute. The Courant number selected for generating results is shown in bold.

Courant Number	Time Steps	Cost [s]	Difference [%]
1.0	1073	0.06	7.01
0.9	1192	0.71	8.03
0.8	1341	0.74	2.99
0.7	1532	0.84	1.98
0.6	1788	0.98	5.03
0.5	2146	1.22	2.00
0.4	2682	1.46	3.02
0.3	3576	1.97	2.02
0.2	5365	2.92	1.01
0.1	10730	5.90	-

TABLE V

Courant number sensitivity study on the concentration of ^{135}Xe with five nuclides in the MSRE based on the time step size used with 500 spatial nodes over the course of one hour. The Courant number selected for generating results is shown in bold.

Courant Number	Time Steps	Cost [s]	Difference [%]
1.0	64384	35.26	0.21
0.9	71538	40.51	0.08
0.8	80481	44.24	0.16
0.7	91978	52.29	0.13
0.6	107308	58.86	0.00
0.5	128769	70.45	0.08
0.4	160962	89.20	0.05
0.3	214616	119.76	0.00
0.2	321924	179.72	0.00
0.1	643848	349.55	-

9. Page 11-line 26: What does "CFL" stand for? I assume it is for "Courant-Friedrichs-Lewy." However, it should be defined earlier in the text.

Solution: Thank you for catching this, the text previously had a longer discussion of the CFL condition but it was removed.

The sentence has been changed to "This sensitivity study investigates the time step size represented by the Courant number, used in the Courant–Friedrichs–Lewy convergence condition."

10. Fig. 1a and Fig. 1b should be explained deeper within the text to clarify the importance of Courant number.

Solution: Modified the paragraph discussing Courant number to provide more discussion on the effect of the Courant number on the results:

In Figure 1(a), the Courant number of 0.1 has an accurate phase, but is highly diffusive. Because the purpose of only using advection is to capture the difference between models with no mixing and perfect mixing, this model is not useful due to mixing by numerical diffusion. Figure 1(b), the Courant number of 1.0 has a slightly inaccurate phase, but does not have numerical diffusion. However, the phase shift can be accounted for by scaling the salt advection speed by a constant, which removes this drawback, as shown in Figure 1(b). For this work, the advection speed was found to be most accurate when multiplied by a factor of 1.002 for each case studied. This means a Courant number of one is optimal for the analyses studied.

11. In Section III.D, there is a code-to-code benchmarking with a prior study which implements nine-region and two-region methods. It is recommended to add a brief explanation of those methods to provide reader a clear picture of the comparison.

Solution: Thank you for noticing this. There is now a more detailed explanation of the code that is being compared:

Comparison between ^{135}Xe chain concentrations are made with a steady-state radiochemical transport model for additional code-to-code benchmarking [6]. The steady-state model uses in-house Python scripts and the same starting equation shown in Equation (5). The MSRE is discretized into either nine or two regions, with flow paths between them to model the salt flow path. The scripts sets the derivative of the concentration with respect to time equal to zero, turning the coupled system of PDEs into a system of ODEs, with one ODE per region.

12. On page 16, the end of the last paragraph of Section III.D, authors repeated almost all the percent differences listed in Table VI. It is unnecessary to include all the number in the text when they are also listed in a table. Instead, it is recommended to state the importance of those differences and a conclusion that can be made from the results.

Solution: This is a good point, repeating data from the table did not provide additional context. Here is the rewritten paragraph:

Table VI shows that there is good agreement between the spatially resolved model and scaled flux model for all nuclides tracked, with the largest difference being approximately $10^{-3}\%$ for ^{135}Xe . When comparing with the two and nine region models, in every case the nine region model has a smaller difference. This is because the level of spatial refinement is finer in the nine region case, which brings it closer to the more spatially refined model presented in this work, which uses 500 spatial nodes. Overall, there is reasonable agreement between the steady-state models and the current time-dependent models. The nine region model, aside from ^{135m}Xe , has differences less than 7% from the spatially resolved and scaled flux models. The larger differences in ^{135m}Xe are expected due to the differing treatment of that specific nuclide.

13. In Section III.E.2, there are some results for Molten Salt Breeder Reactor (MSBR) to give an idea on how results may differ with larger removal rates. However, the information given for this analysis is very limited. It is recommended to add more information for MSBR results, maybe some plots to compare with MSRE, or a separate section reserved for the applications with larger removal rates.

Solution: The reference to the MSBR in this section was intended only to serve as an example for when the reader may see chemical removal rates that are larger and correspond to greater inaccuracy in the scaled flux method. The MSBR itself was not modeled, only varying chemical removal rates in the MSRE (which demonstrate that if a removal rate similar to the MSBR was present, then the scaled flux method would have roughly 4% inaccuracy after one day of simulation.) Thank you for bringing this to light, as this can be made more clear to the reader. This is fixed by removing the initial mention of the MSBR, and instead adding the following into the paragraph discussing the MSRE removal rate:

Based on the xenon removal rate in the MSRE of $4.067 \times 10^{-5} \text{ s}^{-1}$, the difference metric is negligible, meaning the scaled flux model can be used with high accuracy. If the MSRE had a larger chemical removal rate of xenon, such as a rate similar to the rate in the Molten Salt Breeder Reactor (MSBR) of 0.05 s^{-1} , then the difference metric would be non-negligible [19]. For the xenon removal rate of 0.05 s^{-1} , the difference peaks at 7.5% and is at equilibrium at 0.9%. This means that the MSBR, or reactors with similar large removal rates, may need spatial resolution during depletion simulations to accurately model nuclide concentrations.

14. The conclusion section should include more information on the importance of the major findings and the contribution to the literature for future reference.

Solution: Added sentences to conclusion covering importance of the findings and tying results back in with the literature:

The results in this work demonstrate the importance of both the simpler, approximate approaches and the more sophisticated, high-fidelity approaches to spatial resolution of nuclides in MSRs. The scaled flux model generally captures the physics of MSR nuclide

transmutation, and for most cases offers a fast and simple approach that has negligible errors [5]. For example, the MSRE operation from January 24th, 1966 to November 7th, 1967 shows that the scaled flux model has good agreement with MCNP/ADDER and the spatially resolved model [7]. Additionally, for a steady-state solve using constant power and flow rate in the MSRE, the scaled flux method does not introduce significant error [6]. However, for some scenarios previously described, the scaled flux model becomes inaccurate, and the higher-fidelity models are necessary to accurately track nuclide concentrations [9]. Added sentence to conclusion on different aspects of the literature this work can contribute towards:

These nuclide concentrations affect many aspects of MSR operation, including refuel rates [21, 22], reactivity response [23], radio-toxicity of the fuel inventory [24], and more [25].

Reviewer 2

1. Figure 2 is hard to read. It appears that all modeling approaches show identical results, but the experimental data differs. The latter is no problem as it is adequately discussed in the text. However, the identical modeling results shown in the figure are commented on as "reasonable agreement." Additionally, the blue vs. orange line, spatially resolved/scaled flux, appears overlapping, which is only detectable by careful inspection of the figure. Both issues could be addressed by adding a clarifying sentence in the accompanying prose.

Solution: Thank you for pointing this out. The results indeed overlap on this scale, as the actual difference is very small (non-oscillatory flux, no chemical removal, and constant flow rate of 21.75 centimeters per second). The paragraph discussing the figure has been updated to include this:

Figure 2 shows the concentrations of ^{95}Zr and ^{95}Nb over time. In the figure, comparisons are made between the experimental results of the MSRE [15], analysis using MCNP/ADDER [8], and the current work using the spatially resolved and scaled flux models. There appears to be reasonable agreement between the current work and the results from MCNP/ADDER. This agreement shows that the six to seven nuclides are sufficient to get a reasonable concentration for a specific target nuclide compared to a full depletion matrix. There is also agreement between the spatially resolved and scaled flux concentrations. Both models are overlapping on these plots because the simulation of the MSRE during this time frame does not have any properties which induce error in the scaled flux method. This is discussed in more detail in Section III.E.

References

- [1] Office of Nuclear Energy. Energy department's advanced reactor demonstration program awards \$30 million in initial funding for risk reduction projects. <https://www.energy.gov/ne/articles/energy-departments-advanced-reactor-demonstration-program-awards-30-million-initial>, December 2020.